Requirements Engineering & UML Use-Case Modelling

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No. 48 – Incident Escalation & On-Call Rotation Engine

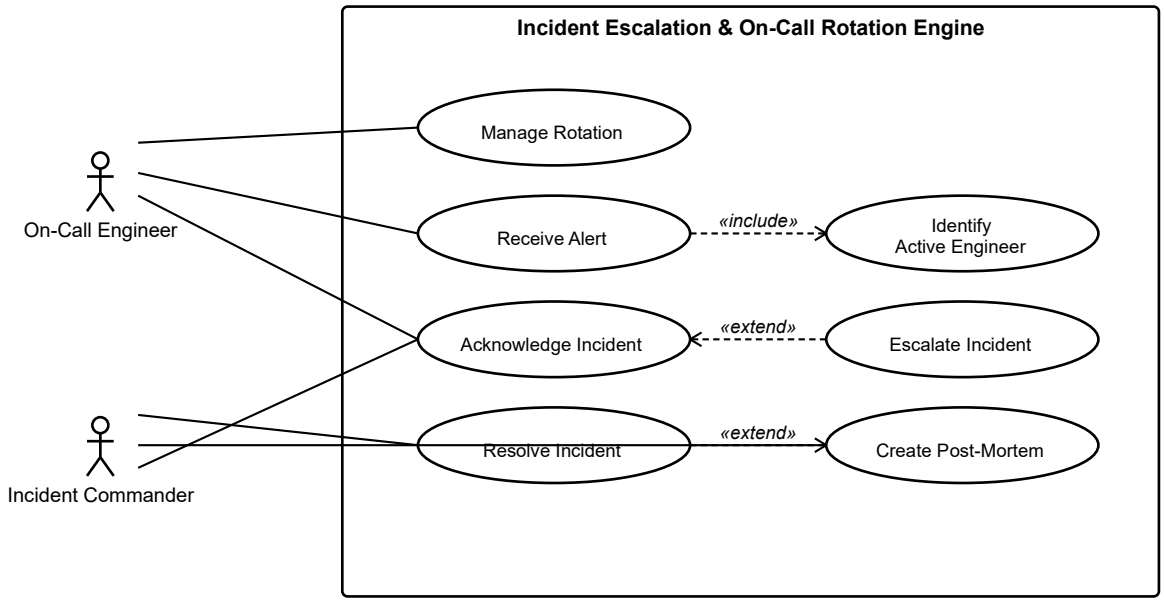
I. Functional & Non-functional Requirements

ID	Description	Priority	Acceptance Criteria	Rationale
FR-001	The system shall allow the Incident Commander to create and manage on-call rotation schedules by assigning engineers to primary and secondary shifts.	High	Pass: A rotation schedule correctly identifies the primary and secondary on-call engineer for every configured shift. Fail: A shift has no valid engineer assignment or overlapping assignments cause an ambiguous active engineer.	Accurate rotations are the foundation of incident routing; an alert cannot be handled correctly without knowing who is currently responsible.
FR-002	The system shall ingest incident alerts from monitoring sources and create an incident record containing the alert details, severity, source, and timestamp.	High	Pass: A valid inbound alert creates an incident record with all required metadata. Fail: A valid alert is dropped or an incident record is created without essential alert information.	Reliable alert ingestion ensures that production incidents enter the escalation workflow and can be tracked from detection to resolution.
FR-003	The system shall identify the active primary on-call engineer from the rotation schedule and dispatch the incident notification through the configured communication channels.	High	Pass: The notification is sent to the engineer assigned to the active primary shift. Fail: The alert is routed to an engineer who is not active or is not dispatched.	Correct routing reduces response time and ensures the person currently responsible is contacted first.
FR-004	The system shall escalate an unacknowledged incident to the secondary on-call engineer when the configured acknowledgement timeout expires.	High	Pass: If the primary engineer does not acknowledge within the timeout, the secondary engineer receives an escalation notification. Fail: The timeout expires and no escalation occurs.	Automated escalation prevents critical incidents from remaining unattended when the primary engineer is unavailable.
FR-005	The system shall allow the Incident Commander to record, review, and track incident post-mortems after an incident is resolved.	Medium	Pass: A post-mortem can be linked to a resolved incident and records the cause, impact, timeline, and follow-up actions. Fail: The post-mortem cannot be associated with the incident or required details are not retained.	Post-mortems help teams learn from incidents, track corrective actions, and reduce the likelihood of recurrence.

II. Non-functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	The system shall initiate alert dispatch through supported channels within 3 seconds of alert ingestion and shall protect incident and user data using secure authentication and encryption in transit.	High	Pass: Performance tests confirm alert dispatch begins within 3 seconds under simulated peak load, and security tests confirm protected authentication and encrypted network communication. Fail: Dispatch exceeds the target or sensitive traffic is transmitted insecurely.	Incident response is time-critical, and operational data must be protected against unauthorized access or interception.
NFR-002	Reliability & Availability	The system shall reliably preserve incident state, escalation history, acknowledgements, and rotation schedules, and shall remain available for critical alert handling during normal infrastructure failures.	High	Pass: Failure and recovery tests show no loss of recorded incident or escalation history, and alert handling resumes without manual reconstruction after a recoverable service interruption. Fail: Incident data is lost or critical alerts cannot be processed after recovery.	The system itself is part of the incident-response path, so data loss or prolonged unavailability can directly increase operational impact.

III. UML Use-Case Diagram



The diagram models the two stakeholders identified in the problem statement: the On-Call Engineer and the Incident Commander. It includes an **«include»** relationship for identifying the active engineer during alert processing and **«extend»** relationships for conditional escalation and post-mortem creation.

IV. Use-Case Flow Specification

Use Case: Escalate Incident

Primary Actor: Incident Commander

Preconditions: An incident alert has been ingested; a valid on-call rotation exists; the active primary and secondary engineers are identified; the incident has not been acknowledged by the primary engineer before the configured timeout.

Main Success Scenario:

1. The system receives and records an incident alert.
2. The system identifies the active primary on-call engineer from the current rotation schedule.
3. The system dispatches an incident notification to the primary engineer.
4. The system waits for acknowledgement until the configured timeout expires.
5. The system detects that the incident remains unacknowledged.
6. The system identifies the secondary on-call engineer for the active rotation.
7. The system escalates the incident and dispatches a notification to the secondary engineer.
8. The secondary engineer acknowledges the incident.
9. The system records the escalation and acknowledgement in the incident history.
10. The Incident Commander can review the current incident status and escalation trail.

Postconditions: The incident escalation is recorded; the secondary engineer has been notified and/or has acknowledged the incident; the incident history contains the notification, timeout, escalation, and acknowledgement events.

Alternate Flow — Primary Engineer Acknowledges Before Timeout:

4a. If the primary on-call engineer acknowledges the incident before the configured timeout expires, the system records the acknowledgement.

4b. The escalation timer is cancelled and the secondary engineer is not notified.

4c. The flow continues with the incident being handled by the primary engineer and the Incident Commander may monitor its status.

End of Submission